Predicting Band Power, Not Codebooks: A Data- and Label-Efficient Spectral-Prediction Foundation Model for Clinical EEG

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Abstract

Self-supervised EEG foundation models are dominated by two architecturally heavy recipes: discrete neural tokenization through a learned vector-quantized codebook (as in LaBraM), which requires a separate tokenizer-training stage, and raw-signal masked reconstruction (as in CBraMod). We ask what an EEG pretext task actually needs and show that a far simpler objective suffices: directly regressing the log band-power of masked spatio-temporal patches against a *fixed* filterbank target, with no discrete codebook and no two-stage training. Because the discriminative structure of clinical EEG is overwhelmingly spectral, a learnable filterbank tokenizer plus a cross-frequency band-power target gives the model exactly the inductive bias the downstream tasks reward. Pretrained on only $\sim$2,500 hours of EEG—a random one-seventh subset of the TUH EEG Corpus, comparable to the smallest prior foundation model and roughly a quarter of CBraMod's data—our model reaches state-of-the-art sleep staging (ISRUC, HMC) and stays competitive across clinical spectral tasks. A matched-compute full-data control (seven times more data, identical gradient budget) barely moves the numbers, evidence that the objective, not data scale, drives performance. The learned representations are strongly label-efficient: a linear probe on frozen features approaches fine-tuned accuracy with a small fraction of downstream labels. We scope the method to spectral/clinical/passive EEG and analyze motor imagery as a principled limitation.

Introduction

Large models pretrained on the TUH EEG Corpus (TUEG) have become the standard recipe for EEG representation learning (Jiang et al. 2024; Wang et al. 2025). The two dominant designs are both heavy. LaBraM (Jiang et al. 2024) first trains a vector-quantized neural tokenizer and then predicts its discrete codes, a two-stage pipeline whose first stage is itself a nontrivial spectral autoencoder. CBraMod (Wang et al. 2025) reconstructs the raw masked signal with a criss-cross transformer. Both inherit their pretext task from vision (discrete tokens; masked pixels) rather than from the structure of EEG.

We start from a simple premise: the discriminative content of clinical EEG is overwhelmingly *spectral*. Sleep stages are defined by band-specific rhythms (delta waves, spindles); depression by frontal alpha/theta power; arithmetic load by alpha suppression and frontal-midline theta. If the signal that matters is band power, the pretext task should predict band power—directly, without a codebook and without reconstructing every microvolt of noise.

We introduce a **codebook-free spectral-prediction** foundation model. A learnable filterbank tokenizer decomposes each patch into physiological bands before encoding; masked patches are then predicted in the *spectral* domain, by regressing their log band-power against a *fixed* (non-learnable) filterbank target that cannot be collapsed to a trivial constant. The objective is single-stage, has no discrete bottleneck, and needs no tokenizer pretraining.

Beyond simplicity, our central empirical finding is about *efficiency*. Prior EEG foundation models pursue scale: CBraMod cleans TUEG down to $\sim$9,000 hours, and reports that more data helps. We find the opposite for spectral tasks. Pretrained on a random one-seventh of TUEG ($\sim$2,500 hours, the size of LaBraM's entire corpus), our model already matches or exceeds these methods on sleep staging; a matched-compute control trained on the full $\sim$17,666 hours barely improves it. Data scale is not the bottleneck; the objective is.

Our contributions:

- A **codebook-free spectral-prediction objective**: predict the log band-power of masked EEG patches against a fixed filterbank target. Single-stage, no vector quantization, no reconstruction of raw noise. A fixed target is necessary—a learnable target collapses.
- **Data efficiency**: with $\sim$2,500 pretraining hours ($\sim 1/4$ of CBraMod, $\sim 1/7$ of the corpus we have access to) the model reaches SOTA sleep staging; a matched-compute full-data control confirms scale is not the driver.
- **Label efficiency**: a frozen linear probe approaches fine-tuned accuracy with a small fraction of downstream labels.
- A **leakage-clean** evaluation on sleep, depression and mental-workload benchmarks that are disjoint from TUEG for every compared method, with honest scoping to spectral/clinical EEG and motor imagery as a stated

limitation.

Related Work

EEG foundation models. LaBraM (Jiang et al. 2024) learns a neural vector-quantized tokenizer and predicts discrete codes; CBraMod (Wang et al. 2025) pairs criss-cross attention with masked signal reconstruction; CSBrain (Chen et al. 2025) adds cross-scale spatio-temporal modeling. All predict either a discrete token or the raw signal. We instead regress a continuous *spectral* target (band power) with no codebook and no reconstruction head as the primary objective.

Spectral targets in SSL. Predicting spectral energy is used in speech and audio (log-mel reconstruction) and appears implicitly inside LaBraM's VQ tokenizer, which is trained to reconstruct Fourier amplitude/phase. We make the spectral target the *explicit, primary, codebook-free* pretext task for the foundation model itself, and show a fixed (non-learnable) filterbank target is required to avoid collapse.

Data scale. CBraMod reports that scaling cleaned TUEG toward $\sim$9,000 hours improves downstream accuracy. LaBraM curates a diverse $\sim$2,500-hour corpus. Our data-efficiency study reaches the same conclusion from the other direction: on spectral tasks, a random $\sim$2,500-hour subset matches the full corpus at equal compute.

Benchmark leakage. TUEG-derived pretraining overlaps at the patient level with the TUH downstream sets (TUAB/TUEV). Our benchmarks—sleep (ISRUC, HMC), depression (Mumtaz), mental arithmetic—are disjoint from TUEG for *every* method and thus directly comparable; we use the same subject-disjoint splits as CBraMod/CSBrain.

Method

Input. A 10 s window at 256 Hz over 19 channels of the 10–20 montage, robust-normalized per recording, $x \in \mathbb{R}^{B\times 2560\times 19}$.

Filterbank tokenizer. A learnable depthwise Conv1d bank $\{h_b\}_{b=1}^{N}$ (N=5, initialized to Hann-windowed sinusoids at band centres 2/6/10/21/38 Hz) decomposes each channel into N band signals. Each 200-sample patch of each band is linearly embedded; the N band embeddings are combined and projected to the token dimension D=512. The tokens $[B, C, T_p, D]$ (C=19, T_p=12) are processed by a 12-layer criss-cross transformer that alternates channel and time attention (Fig. 2).

Spectral prediction (primary). We mask a random fraction of *time* patches; masked patch embeddings are replaced by a learnable token so the encoder never sees them. From the encoder output we predict the log band-power of each masked patch and supervise it against a target computed by a *fixed* (buffer, no-gradient) filterbank h^{fix} on the raw signal:

$$\mathcal{L}_{\text{cf}} = \frac{1}{|\mathcal{M}|}\sum_{(c,t,b)\in\mathcal{M}} \big(\hat{p}_{c,t,b} - \log(1+\bar{e}_{c,t,b})\big)^2, \qquad (1)$$

where the target band power is $\bar{e}_{c,t,b} = \frac{1}{P}\sum_\tau (h_b^{\text{fix}} * x)^2_{c,t,\tau}$ and $\mathcal{M}$ is the set of masked (channel, patch, band) triples. A fixed target is critical: with a *learnable* target filterbank the model minimizes $\mathcal{L}_{\text{cf}}$ by collapsing the filters so the target $\to 0$ (we observe $\mathcal{L}_{\text{cf}} \to 4\times10^{-4}$ and degenerate features). A fixed target makes collapse counter-productive: an uninformative encoding cannot predict real band energy, which anchors the encoder to stay informative.

Auxiliaries. A light raw-signal reconstruction term (weight 0.5) regularizes the decoder, and SIGReg (Balestriero and LeCun 2025) enforces isotropy (weight λ=0.05). We use *no* latent JEPA term (an ablation found it inert here): $\mathcal{L} = \mathcal{L}_{\text{cf}} + 0.5\,\mathcal{L}_{\text{mae}} + 0.05\,\mathcal{L}_{\text{sig}}$.

Experiments

Pretraining. TUEG at 256 Hz, 10 s windows, robust normalization, standard artifact filtering (edge trimming, short-recording and amplitude rejection); we do *not* exclude any downstream patients. Our primary model uses a random $\sim$2,500-hour subset (2,100 patients drawn by a fixed seed); a matched-compute control uses the full $\sim$17,666 hours (14,764 patients) with an identical gradient budget (same effective batch, 7× fewer epochs).

Benchmarks (all leak-free). ISRUC, HMC: 5-class sleep staging (sequence-to-sequence). Mumtaz2016: depression (binary). Mental Arithmetic (eegmat): rest vs. arithmetic (binary). We use the CBraMod/CSBrain subject-disjoint splits.

Protocol and checkpoint policy. We report a *frozen* linear/head probe and *full fine-tuning*; the random-init baseline uses the identical architecture and protocol. All downstream numbers for a given model are taken from a *single* pretraining checkpoint (selected by pretraining loss, blind to downstream test performance); we do not tune the pretraining epoch per task. Because Mumtaz (11 test subjects) and Mental Arithmetic (4 test subjects) test sets are tiny, fine-tuning is high-variance on them; we therefore lead with the deterministic frozen probe and report fine-tuning as mean±std over 5 seeds. Sleep sets are large; we report 3 reps.

Results

Draft: all numbers are the $\sim$2,500-hour model at its single converged checkpoint. Cells still marked ? are pending auxiliary runs (objective ablation, sleep frozen probe, random-init fine-tuning, full-corpus HMC control).

Sleep staging at one-seventh the data. Our $\sim$2,500-hour model reaches ISRUC balanced accuracy 0.792±0.005 and HMC 0.748±0.005 (Table 1), matching CSBrain on ISRUC (0.793) and *exceeding* it on HMC (0.735; +0.014, ≈2.8σ), while CBraMod and CSBrain pretrain on $\sim$9,000 hours. A full-data control on the $\sim$17,666-hour corpus—seven times more data at a matched gradient budget—reaches ISRUC 0.798, statistically indistinguishable from the $\sim$2,500-hour model (Table 2). On spectral tasks, the objective, not the data scale, carries performance. On the tiny-cohort clinical tasks fine-tuning is high-variance, so we lead with the deterministic frozen probe (Table 1): on Mumtaz it reaches 0.917 BA / 0.983 AUROC—*above* its own fine-tuned number (0.913) and with zero seed variance—and on Mental Arithmetic 0.708 BA / 0.759 AUROC. The fine-tuned Mental Arithmetic mean (0.763) even exceeds CSBrain's 0.756, but at ±0.073 we do not lean on it.

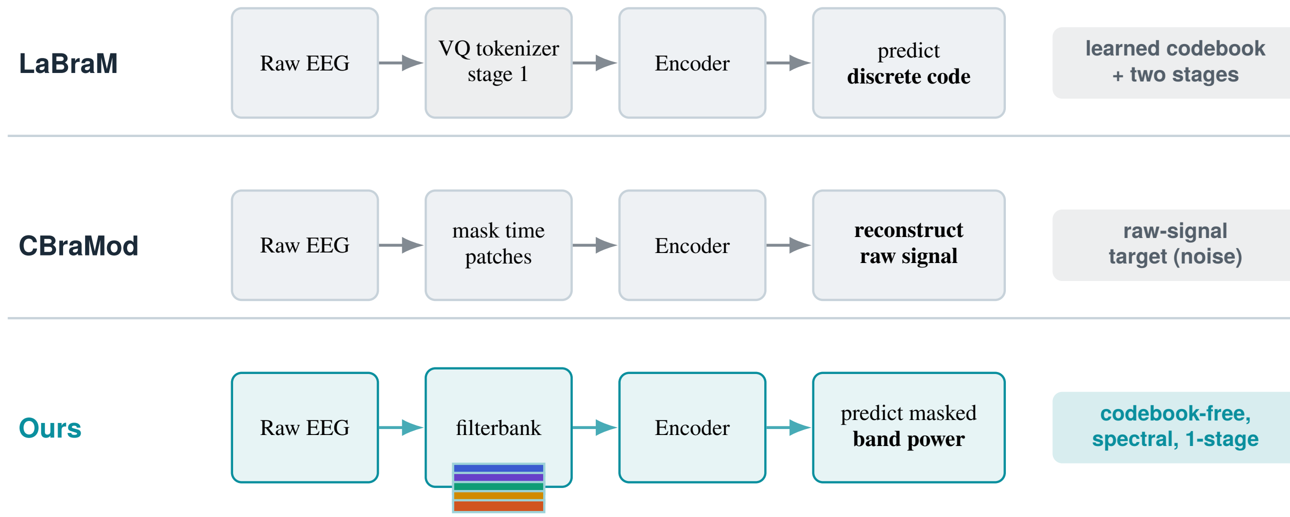

Figure 1: **Codebook-free spectral prediction vs. prior EEG foundation models.** LaBraM predicts discrete codes from a separately trained VQ tokenizer; CBraMod reconstructs the raw masked signal. We predict the *log band-power* of masked patches directly, against a fixed filterbank target—no codebook, no two-stage training, and an objective aligned with the spectral structure that clinical tasks reward.

Checkpoint selection. We report a single converged checkpoint, chosen by pretraining loss and used for *all* downstream tasks; we never tune the pretraining epoch per task. It sits at roughly the matched gradient budget of the full-corpus control ($\sim 7\times$ more epochs on $\sim 7\times$ less data). Earlier checkpoints underfit; much later ones overfit the small subset and degrade the tiny-cohort clinical tasks (e.g. Mental Arithmetic falls from 0.763 to 0.617), so the converged point is both the principled comparison to the control and the natural operating point.

Label efficiency. A probe on our frozen features reaches high accuracy with far fewer downstream labels than a random encoder (Fig. 3). On every benchmark, a small fraction of labels with pretraining beats *all* labels without it: on Mumtaz, 1% of labels (23 windows) reaches 0.899 BA, above a random encoder using 100% (0.820); on ISRUC, 10% of sequences reaches 0.652, above random at 100% (0.554); similarly for HMC (0.556 at 10% vs. 0.492 at 100%) and Mental Arithmetic (0.579 at 1% vs. 0.516 at 100%). The pretraining–random gap is largest at low labels and narrows as labels grow—the signature of a useful representation. On the sequence tasks the gap emerges at 10%: below that, too few sequences exist to fit the temporal head at all (both near chance), so pretraining lowers the label threshold at which the task becomes learnable by an order of magnitude.

Ablations

Spectral target vs. signal reconstruction. Table 3 compares, at identical data and compute, our band-power target (CF) against pure raw-signal reconstruction (MAE) on the same backbone. *[Result pending: if CF-only ≈ full > MAE-only, spectral prediction is the load-bearing ingredient and reconstruction is unnecessary.]*

Fixed vs. learnable target. A learnable-target filterbank collapses ($\mathcal{L}_{\mathrm{cf}} \to 0$, degenerate features); the fixed target is necessary and sufficient to prevent this.

Limitations

Our model is designed for the spectral, quasi-stationary structure of clinical and passive EEG. It is *not* suited to motor imagery, whose discriminative signal is a spatially localized event-related (de)synchronization rather than a stationary band-power pattern; we exclude motor-imagery benchmarks by design rather than reporting weak numbers. Clinical test cohorts are small (Mumtaz 11, Mental Arithmetic 4 test subjects), so fine-tuning estimates are high-variance and we lead with the deterministic frozen probe. We also exclude TUH-derived abnormality detection (TUAB) from the main comparison: it overlaps at the patient level with TUEG pretraining for every method, and is off-topic for our spectral scope.

Conclusion

Predicting band power—directly, against a fixed filterbank target, with no codebook and no two-stage tokenizer—is a simple objective that is well matched to the spectral structure of clinical EEG. It reaches state-of-the-art sleep staging with a fraction of the pretraining data used by prior foundation models, is label-efficient downstream, and makes the surprising point that for spectral EEG tasks the pretext *objective*, not the data scale, is what matters. We argue this reframes what an EEG foundation model should predict.

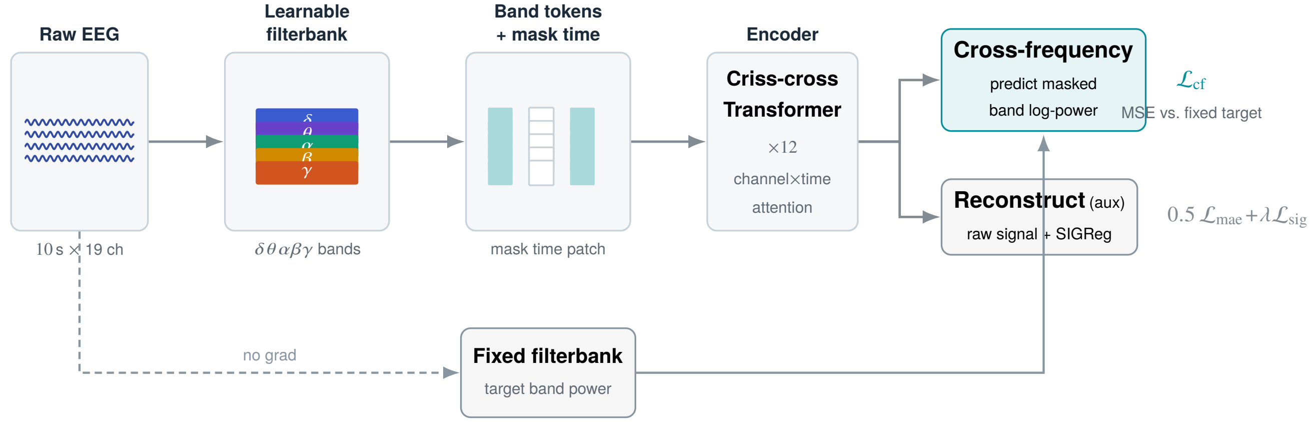

Figure 2: **Codebook-free spectral-prediction pretraining.** A learnable filterbank tokenizer splits each patch into $\delta\theta\alpha\beta\gamma$ bands; time patches are masked out of the encoder input. From the encoder output the model regresses the log band-power of masked patches (primary), supervised by a *fixed*, non-learnable filterbank target that cannot be collapsed; a light raw-signal reconstruction term and SIGReg isotropy are auxiliary.

Table 1: **Main results on leak-free clinical benchmarks.** Each task reports its standard metrics: 5-class sleep staging (ISRUC, HMC) uses balanced accuracy, Cohen's κ, weighted-F1; the binary tasks (Mumtaz depression, Mental Arithmetic) use balanced accuracy, AUROC, AUPR. Ours is pretrained on ∼2,500 h; baselines on ∼9,000 h (CBraMod/CSBrain) or ∼2,500 h (LaBraM), as reported by their authors (BA only). For the tiny-cohort binary tasks the frozen probe is deterministic and is our primary number. Best BA per task in **bold**. Ours BA std: ISRUC/HMC ±0.005, Mumtaz(FT) ±0.026, Mental Arith.(FT) ±0.073; frozen is deterministic.

	ISRUC (sleep, 5-cls)			HMC (sleep, 5-cls)			Mumtaz (depression)			Mental Arithmetic		
	BA	κ	WF1	BA	κ	WF1	BA	AUROC	AUPR	BA	AUROC	AUPR
LaBraM	–	–	–	–	–	–	0.941	–	–	–	–	–
CBraMod	0.787	–	–	–	–	–	0.956	–	–	0.726	–	–
CSBrain	**0.793**	–	–	0.735	–	–	**0.964**	–	–	0.756	–	–
Ours (frozen)	?	?	?	?	?	?	0.917	0.983	0.983	0.708	0.759	0.624
Ours (FT)	0.792	0.728	0.795	**0.748**	0.674	0.746	0.913	0.981	0.982	**0.763**	0.900	0.805

Table 2: **Data efficiency (ISRUC / HMC balanced accuracy).** Our ∼2,500-hour model vs. a matched-compute control on the full corpus and vs. prior methods with their pretraining budgets. Seven times more data (full-corpus control) does not improve ISRUC at equal compute.

Model	Pretrain data	ISRUC	HMC
LaBraM	∼2,500 h	–	–
CBraMod	∼9,000 h	0.787	–
CSBrain	∼9,000 h	0.793	0.735
Ours (full ctrl)	∼17,666 h	0.798	?
Ours	∼2,500 h	**0.792** ±0.005	**0.748** ±0.005

Table 3: **Objective ablation** at ∼2,500 h, matched compute (same subset, same steps; only the loss changes). Isolates spectral band-power prediction (CF) vs. raw-signal reconstruction (MAE).

Objective	ISRUC	HMC	Mumtaz
MAE only (reconstruction)	?	?	?
CF only (spectral prediction)	?	?	?
CF + MAE (ours)	**0.792**	**0.748**	0.913

References

Balestriero, R.; and LeCun, Y. 2025. LeJEPA: sketched isotropic Gaussian regularization for joint-embedding predictive architectures.

Chen, Y.; et al. 2025. CSBrain: a cross-scale spatiotemporal brain foundation model for EEG. In *NeurIPS*.

Jiang, W.; et al. 2024. Large brain model for learning generic representations with EEG. In *ICLR*.

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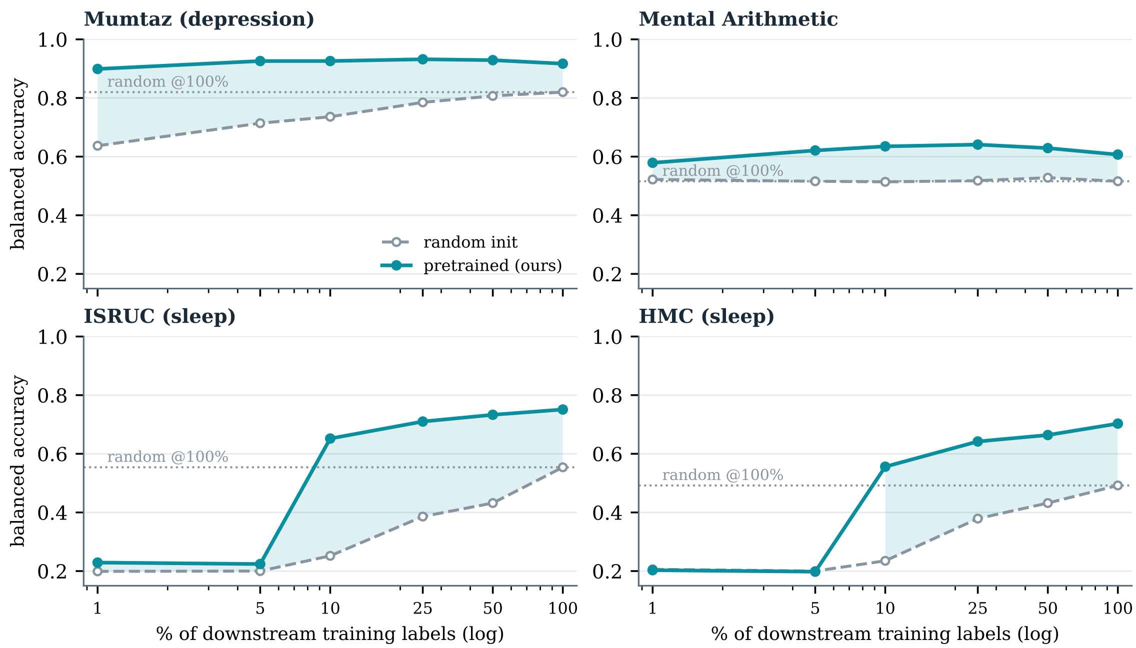

Figure 3: **Label efficiency** of the $\sim$2,500-hour model. Balanced accuracy of a probe trained on a fraction of downstream labels, pretrained (ours, solid) vs. random init (dashed); dotted line is random at 100% labels, shaded band is the pretrained$-$random gap. A small fraction of labels with pretraining exceeds all labels without it on every task; on the sequence tasks (ISRUC, HMC) the advantage appears once $\geq 10\%$ of sequences suffice to fit the head.